

Image Analytics for Train Crowd Estimation

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Abstract— Overcrowding is a common problem faced by train commuters in many countries. While waiting for the train at the stations, commuters tend to cluster and queue at doors that are closest to escalators and elevators that lead towards the station entrances and exits. This scenario results in trains not being fully utilized in terms of their capacity. As cabins with certain door positions tend to be more crowded than the rest of the cabins. The objective of this paper is to provide a methodology to estimate the crowd density within cabins of incoming trains, while leveraging on the existing train CCTV infrastructures. Providing the train cabin density information to commuters who are waiting for the incoming train allows the commuters to better select which cabin to board based on the provided density information. This will facilitate a better commuting experience without incurring a high cost for the train operator. To achieve this objective, we have adopted the usage of deep convolutional neural networks to analyze the footage from the existing security camera inside the trains and classify the images frames based the crowd level of train cabins. Three different experiments were conducted to train and test different convolutional neural network models. All models are able to make classification with an accuracy rate of over 90%.

I. INTRODUCTION

In year 2016, 54% of the world's population lived in urban areas. This number is expected to reach 60% by 2030 [1]. Fast growing population in a city causes overcrowding in public transports and the crowd density is becoming more difficult to manage during peak hours. Larger population and more developed infrastructures will not only lead to prosperity, but also to a busier transport network. Therefore, longer waiting time and the inability to board public transports such as the train are to be expected.

Fig. 1. shows a survey result on Customer Satisfaction Index of Singapore [2]. This survey was targeted on Singapore Mass Rapid Transit (MRT) system. The satisfaction score fell 1.6 points compared to previous year. One of the main reason for dissatisfaction is the crowd management. Although manpower were deployed to handle the crowd at train platforms, crowd density inside the train cabin is still unpredictable. Also, additional expenditure is incurred for employing such crowd control staffs.

In Singapore, for safety and security purposes, each train is installed with an average of 14 closed-circuit television (CCTV) cameras (2 CCTVs per cabin). These cameras allow the staff to monitor the situation inside the train cabins and take necessary actions in case of emergency. Thus, using the images from the CCTV to estimate crowd density in the train is the most cost-effective way, as it does not require installation of additional sensors or infrastructure.

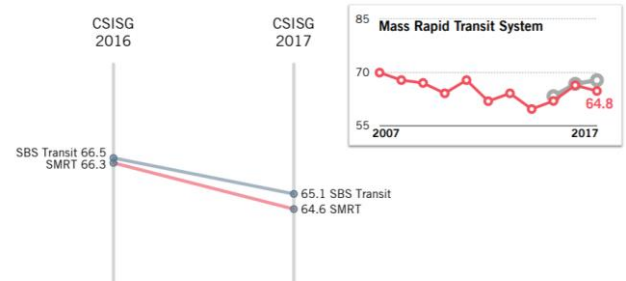


Fig. 1. Statification Score for Train System in 2017

II. RELATED WORK

There has been several attempts at estimating crowd level density. The different attempts used different input data and methods.

A. Passenger Load Sensors

Singapore Land Transport Authority (LTA) launched a new system that displays crowd level information on several LCD screens at train platforms [4]. The LCD screen will display the current capacity of each train cabin using three color codes, where green indicates not crowded, amber indicates slightly crowded with only standing space available and red illustrates limited standing space in the train cabin. The system uses the load sensor on the train to derive the data. The load sensor data is transmitted to the next train station once passengers have alighted and boarded the train. This system guides passenger to less crowded cabins, however, it cost S\$1.5 million and required 8 months to develop. The approach also requires continuous access to the load sensor data, which is difficult to obtain, and requires constant update and maintenance of the load sensors, thus requiring more resources.

B. Infrared Sensors

Another approach recently installed four infrared sensors on buses [5]. Two infrared sensors were placed at the entrance while the other two were placed at the exit path. The sensors will capture the human hindrance and therefore increase or decrease the crowd count. This approach however is not applicable to train crowd density estimation as multiple passengers enter and exit the train doors at the same. The system is unable to handle occlusion [3]. In addition, this approach is less preferred as it would require installation of additional hardware equipment resulting in additional time and resources needed.

C. Image Processing Techniques

Edge detection is an image processing technique that measures the total perimeter of the edges of objects in the image. This technique is used to highlight the silhouette of people standing in the train cabin. The process can be further optimized by thinning the edges. However, this technique is only capable of handling low density crowds. This is due to the method's limitation in handling the overlapping and occlusion of individuals which may cause the error rates to increase and affect the accuracy of the overall classification results [6]. Fig. 2. is an example of edge detection on pedestrian.

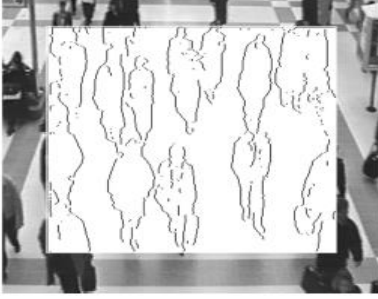


Fig. 2. Edge detection on pedestrian

D. Artificial Neural Network

One of the tools commonly used in machine learning is artificial neural networks. Neural network is an excellent tool as it is able to find patterns and transform the input layer to an output layer that is able to teach the machine to recognize [7]. There are many different types of neural network with its own complexity and use cases, such as feedforward neural network, recurrent neural network, convolutional neural networks, Hopfield networks, Boltzmann machine networks and many more. A recent work [3] used neural network to estimate crowd density, however the approach focused on estimating crowd density for indoor scenarios as their experiments made use of indoor image sets.

This paper will demonstrate the usage of convolutional neural network (CNN) to train a model that is able to classify the crowd density in a train cabin. As seen in [11] - [13], CNN performs well in image classification. CNN adopts a layered approach where features that can recognize and interpret objects easily will be detected on the first layer. Next layer will detect the features that are more abstract and last layer will combine all the detected features into an input data.

III. IMPLEMENTATION

A. System Architecture Overview

Before being able to classify images based on their features, a model has to be derived by feeding the Convolutional Neural Network with existing training images and their corresponding labels. The network also needs to be validated on how well it can classify the train's crowd density using a validation image set. Fig. 3 shows an overview of the system architecture for the training process.

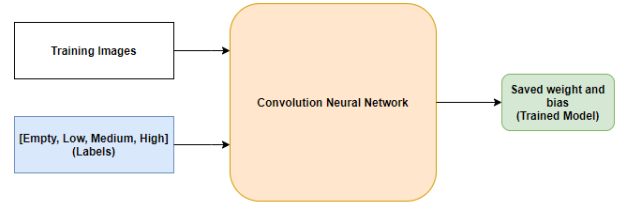


Fig. 3. System Architecture (Training)

After the model has been generated, it is then loaded into the convolutional neural network. The model is then used to make classification on a new and independent test dataset. Fig. 4 shows an overview of the system architecture of the testing process.

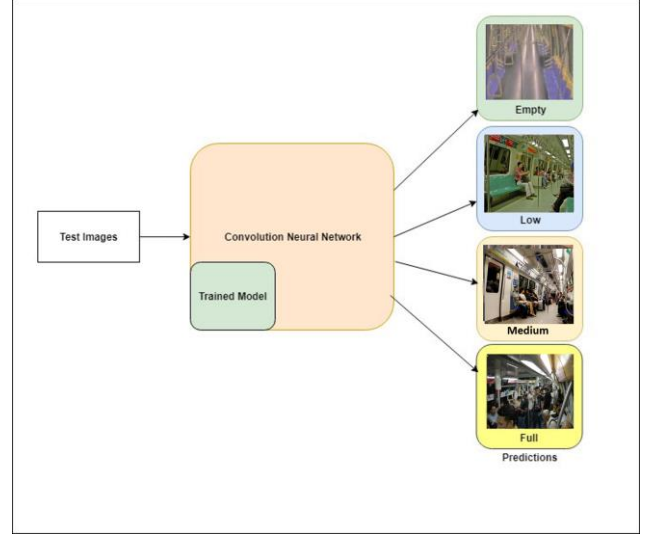


Fig. 4. System Architecture (Testing)

B. Convolutional Neural Network Architecture

Fig. 5 shows the convolutional neural network architecture. This network architecture consists of several layers, these layers help reduce over-fitting, decrease memory usage and speed-up training. More in-depth explanation of each layer will be further elaborated in section C.

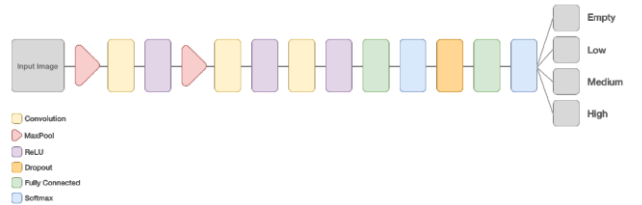


Fig. 5. CNN Architecture

C. Convolutional Neural Network Layers

Convolution Layer

The most important building block of a CNN is the convolution layer. In the convolution layer, the input image is combined with the kernel, which are used to

recognize important features of an image. By moving the kernel over the image, the kernel is centered over the elements of the image every time it moves along the image. The overlapping values are multiplied and the result is accumulated. The value of the kernel and its complexity is derived via back-propagation during the training process and is shared throughout all convolution layers within the CNN. The convolution of both tensors outputs a feature map which is the input for subsequent layers in the network.

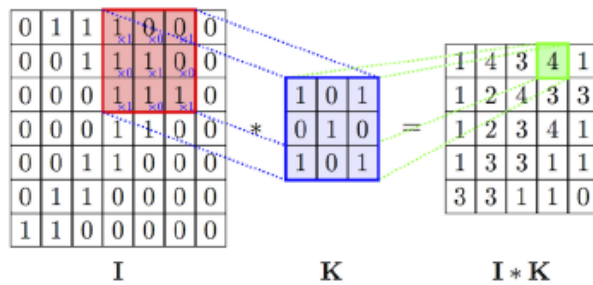


Fig. 6. Convolution Layer [8]

Pooling Layer

The pooling layer helps to reduce overfitting and scales down the size of the input while keeping important information for the subsequent layer, thus improving performance. One of the methods for pooling is Max-Pool. Using a kernel, it strides over the input image and chooses the maximum value found within the kernel size, retaining the most valuable information. In this CNN, a pooling layer was used as the first layer to significantly increase performance.

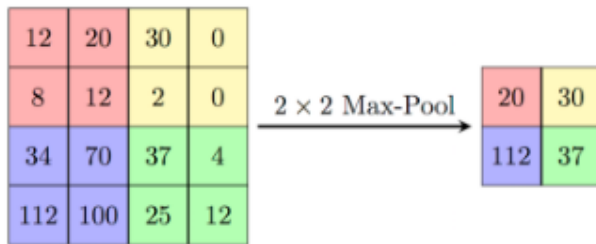


Fig. 7. Pooling Layer [8]

Activation Function

Activation functions are used in combination with the outputs of other layers to generate a feature map. They are used to introduce non-linearity into the network, representing more complex changes in input. This way, it allows the network to train on the complex patterns found in data. Rectified Linear Unit or ReLU activation function is primarily used in CNNs because of its performance. The convolution layers are usually paired with ReLU layers and repeated. A ReLU computes a linear function of the inputs, and outputs the result if it is positive, and 0 otherwise.

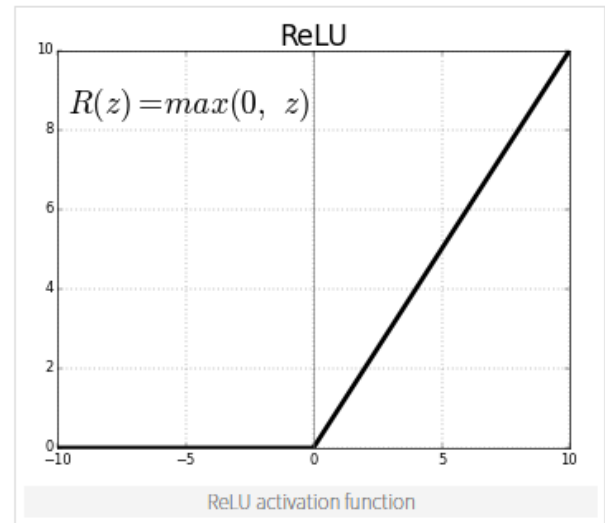


Fig. 8. ReLU activation function [9]

Dropout Layer

The dropout layer is used during training to randomly neurons on the layer to be temporarily “dropped out”, which means it will be entirely ignored during the training process. This prevents overfitting by disallowing nodes from being too reliant on another neighboring node being present. Overfitting occurs when the network memorizes certain features during the training process and only performs well with the training dataset but poorly when used on unseen data. The dropout layer can help the model regularize better with unseen data and avoid overfitting.

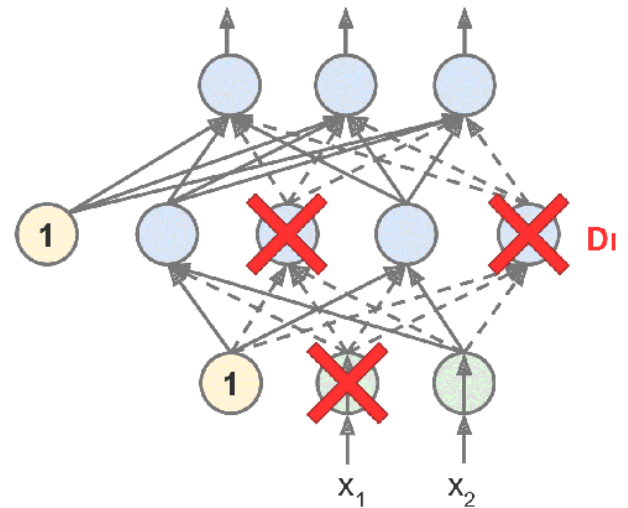


Fig. 9. Drop out regularization [10]

Fully-Connected Layer

The fully-connected layer consist of neurons that are connected to all activations in the previous layer. It essentially captures elements of the input feature map that consist of the features learnt through the network. This allows the fully-connected layer to make a prediction on the image’s classification.

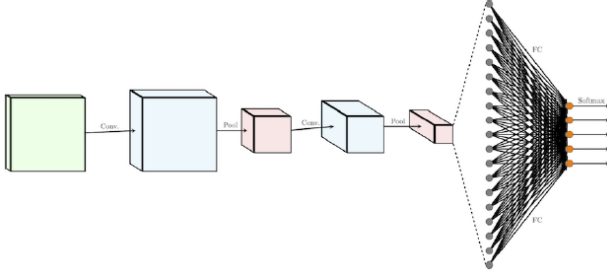


Fig. 10. Fully-Connected Layer [8]

IV. EVALUATION

A. Hyper-Parameters

There are several hyper-parameters that needed to be configured based on testing different values and models to derive the most optimal results.

Epoch

One epoch refers to when an entire dataset is passed forward and backwards through the CNN once. To optimize learning, the dataset has to be passed through the network multiple times. Too low of an epoch will lead to underfitting as the network does not get to learn enough from the dataset while too high leads to overfitting as the network starts to memorize too much information from it. Epoch is set to 10.

Learning Rate

Learning rate refers to the rate of how fast the network will adapt to the new dataset. In the training process, an iterative optimization approach, called gradient descent, is used to tweak the model parameters iteratively in order to minimize loss through a number of training steps, ideally until the algorithm converges to a minimum. Learning rate determines how fast the gradient descent converges, if it is too small, it may result in underfitting as the algorithm will have to go through many iterations to converge. If it is too high, the algorithm may simple “skip over” the minimum and may never find it. Learning rate is set to 0.001.

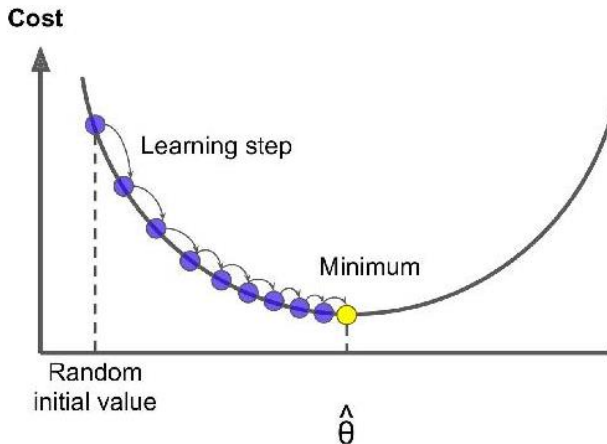


Fig. 11. Gradient Descent [10]

Number of Kernels

Each kernel in a CNN is used to recognize certain feature of an image. The number of kernels intuitively should correspond to the number of features. The number of kernels also corresponds to the number of feature maps which allows the network to learn explanatory features within the image. However, a higher number of kernels may lead to poor detection rate of the network because there are too many features to be “fulfilled”. On the other hand, a low number of kernels may lead to be being able to learn enough features to make predictions. Number of kernels used are 16, 32, and 16.

Kernel Size

Kernel size refers to the width and height of the kernels used within the convolution and pooling layers. A larger kernel enables the network to learn more complex feature but it would take up more computation time. Kernel size used: 7, 7, 5 (convolution) and 2 (pooling).

Stride

As a convolution shifts along the input, it uses the stride parameter to determine how it shifts over the input. A larger image dimensionality would require a large amount of time to scan over every pixel and determine its important feature. It would be optimal to have a larger stride for larger images to skip certain elements instead of going over every pixel. On the other hand, a smaller stride will be useful for smaller images to avoid not detecting important information. Kernel stride used: 1 (convolution) and 2 (pooling).

B. Dataset

A total of 5750 CCTV frame images from SBS Transit, a Singapore public transport company, were used for this experiment. The images were of size 704*576 pixels. Due to privacy concerns, we are only able to show the empty train cabin image, other images in this paper are taken separately for illustration purposes. The goal is to automatically classify the images into one of the four categories: - Empty: the cabin is empty, Low: has empty seats or has a lot of standing space, Medium: no empty seat but still plenty of standing space, High: very little standing space or no standing space. Figure 12 shows an example of an Empty train cabin. The images were manually annotated for groundtruth for training and testing purposes.



Fig. 12. Sample image of an empty train cabin

Table I shows the breakdown of images according to four category.

TABLE I. NUMBER OF IMAGES FOR EACH CATEGORY

Category	Number of Images
Empty	1687
Low	3323
Medium	214
High	526

C. Experiment Setup

Each train cabin is installed with two cameras. Images were collected from four cameras within two cabins for a total of 2-hour train journey. Even though there were two cameras, the angle of the images were not affected by the different placement of camera, as such the images were all combined. Image frames in the dataset are sorted in chronological order based on their timestamp.

The images were further divided into different sets: training, validation and testing. Three different experiment setups were conducted due to the nature of the dataset: Experiment One selected only the first 200 to 300 frames from each category to maintain a balanced dataset. Experiment Two randomly selected 200 to 300 images from each category, while Experiment Three made use of the whole dataset. Experiment One and Two were conducted to see how the models would perform on a balanced dataset. For all experiments, 80% of the images were used for training, 10% for validation and the last 10% for testing. Each experiment is independent of itself and consist of its own training, validation and testing dataset

Table II below shows the breakdown of how the images were used in each the experiments.

TABLE II. DATASET USED

	Total Images Used	Training (80%)	Validation (10%)	Testing (10%)
Experiment One				
Empty	200	160	20	20
Low	200	160	20	20
Medium	210	168	21	21
High	300	240	30	30
Experiment Two				
Empty	200	160	20	20
Low	200	160	20	20
Medium	210	168	21	21
High	300	240	30	30
Experiment Three				
Empty	1680	1344	168	168
Low	3320	2656	332	332
Medium	210	168	21	21
High	520	416	52	52

As the CNN does not accept any raw image for classification, therefore all images were converted into an

image matrix as an array. All the images were also scaled to a height of 64 pixels. This image size was chosen is because it enough to contain valuable information and handling decent low training time [3]. In addition, to prevent any color-bias in the training model, the images were also converted to greyscale.

A common way to label data for categorization of multi-class data is to use one-hot encoding. An array is used to identify the classification of an image based on the index of 1 as shown in Table III. This array is tagged to each respective image.

TABLE III. ONE HOT ENCODING

If	Empty	Low	Medium	High
Then	1	0	0	0
Represents	Empty			

D. Results and Discussion

Table IV, V, and VI show the confusion matrices from each of the experiments. The table shows that the model is able to classify the *Empty* train cabins perfectly. All three models had problems distinguishing *Medium* from *High* images, this is most probably due to the fact that there are standing passengers in both image classes, and the only distinguishing factor is the floor space.

Table VII shows the overall classification accuracy for each of the experiments. The results show that Experiment One had obtained the highest overall average accuracy rate with 98.90%, followed by experiment Three with an overall average accuracy 95.64% and experiment Two with an overall average accuracy of 94.51%. All three experiments show satisfying results as the accuracy rates achieves are over 90%. It is in fact heartening to see the that Experiment Three has a better performance then Experiment Two as it shows the dataset does not need to be balanced to achieve a good overall accuracy rate. This is in fact more representative of real-life scenario, as *High* train density would on average occur less than the other labels as the trains are mostly full only during peak hours, which on average are about 1.5 hours in the morning and 1.5 hours in the evening, out of the 19 hours of daily train operations.

TABLE IV. CONFUSION MATRIX FOR EXPERIMENT ONE

(%)	Empty	Low	Medium	High
Empty	100	0	0	0
Empty	0	100	0	0
Medium	0	0	95.24	4.76
High	0	0	0	100

TABLE V. CONFUSION MATRIX FOR EXPERIMENT TWO

(%)	Empty	Low	Medium	High
Empty	100	0	0	0

Empty	15	85	0	0
Medium	0	0	90.48	9.52
High	0	0	0	100

TABLE VI. CONFUSION MATRIX FOR EXPERIMENT THREE

(%)	Empty	Low	Medium	High
Empty	100	0	0	0
Empty	0.3	99.4	0.3	0
Medium	0	0	100	4.76
High	0	0	44.23	55.77

TABLE VII. OVERALL ACCURACY

Experiment	Overall Acc (%)
One	98.9
Two	94.51
Three	95.63

V. CONCLUSION

Overcrowding is a common problem faced by train commuters in many countries. In this paper, we have proposed a methodology to estimate the crowd density inside train cabins based on the images obtained from the CCTV inside the train cabins. The methodology is based on convolutional neural network to automatically classify the image frames from the CCTV camera. By providing information regarding the crowd density inside the incoming trains to the commuters waiting at the station, the commuters can better disperse themselves on the platform, thus avoiding overcrowding in certain train cabins. This method does not require any changes or additional infrastructure to the train operators, and is very cost-effective. Possible enhancement to the system include building a web application or mobile application that is connected to the back-end convolutional neural network to serve as an application to commuters for easier access.

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